MINOR IN APPLIED PETROLEUM ENGINEERING TECHNOLOGY

Rationale

The Applied Petroleum Engineering Technology (APET) minor addresses the expected national need for engineering personnel knowledgeable in the Oil and gas industries' transportation, processing, and storage. It also prepares undergraduate students for possible graduate studies in petroleum engineering and related fields.

It is primarily geared towards mechanical engineering students, but it should also accommodate students from other engineering backgrounds, provided all prerequisite courses are completed.

The program is primarily concerned with the technical, environmental, and economic aspects of oil and gas extraction as well as downstream processing and handling. The goal is to apply fundamental knowledge of upstream and downstream operations to enhance the graduate's ability to perform effectively as a practitioner in a short period of time.

Program participants are required to complete at least 18 credits of related coursework. Although not required, with proper and timely advising, they may also complete their Co-op work experience or their BS project in Oil and Gas, thereby enhancing their capabilities.

To effectively cover the added coursework in the fall and spring semesters, one full-time faculty member with a specialty in Petroleum/Chemical Engineering is required.

Program Objectives

The primary objectives of the APET minor program are to:

- Give engineering students basic knowledge of the Oil and Gas industry;
- Endow students with the ability to deal with applied Oil and Gas downstream engineering;
- Prepare qualified Lebanese graduates to become technically productive contributors to the future Lebanese petroleum sector;
- Provide an impetus for undergraduate students to pursue graduate studies in the Oil and Gas domain.

Learning Outcomes

After completing the minor program, students should have the ability to:

- Apply relevant knowledge of mathematics, science, and engineering to identify, formulate, and solve practical problems related to downstream petroleum engineering processes;
- Use modern engineering tools relevant to core petroleum engineering practices;
- Deal with the uncertainty and risk involved in petroleum transportation, refining, and storage problems.

Career Options

This minor enables APET holders to begin careers as field technologists in the exploration, transportation, refining, and storage sectors of the conventional oil and gas production industry. Typical employers include oil and gas Companies; oil service companies, including the QA/QC sector; government-related agencies; and petrochemical plants and refineries.

Program Requirements

While most relevant to mechanical engineering students, this minor is offered to undergraduate RHU students in the BS program, from different disciplines (mechanical, mechatronics, civil, electrical, computer, communications, and biomedical). It may also be offered to students with a BS/BE from other universities, subject to a case-by-case evaluation of their transcripts and other specific RHU requirements. Proper advising is pertinent to the timely completion of the minor.

To complete the APET minor, a student must:

1) Declare an APET minor by completing the Minor Declaration Form.
2) Obtain the approval of the dean of the college major and the dean of the College of Business Administration;
3) Obtain a Minor Cumulative Grade Point Average of no less than 70 %.
4) Complete 18 credits of coursework as specified below.

APET Curriculum

The program strikes a proper balance between fundamentals and practice while preparing students to become productive contributors in the oil and gas industry. Accordingly, this minor shall include:

- Oil and gas chemistry and properties
- Design and analysis of systems and procedures for producing and handling fluids;
- Decision-making under conditions of risk and uncertainty;
- Safety hazards and environmental impact consideration and protection;
- Pipeline design and installation;
- Above-ground storage and tanks;
- Gas and oil processing and storage (gas, crude oil, and refined oil products);

- Refinery units and equipment inspection using codes and standards of international organizations such as API and ASME;
- Quality assurance and quality control aspects;
- Non-Destructive Testing (NDT)
- National and regional macroeconomic impact.

The APET curriculum comprises 18 credits: 15 mandatory and 3 elective, selected to satisfy the requirements of the proposed program objectives and learning outcomes. The courses that encompass the minor are given below.

I. Mandatory Courses

Description of the five mandatory courses (15 credits) follows.

APET 200	**Petroleum Engineering**	**3(3,0)**
Overview of the petroleum engineering and petroleum industry: Nature of oil and gas reservoirs, exploration and drilling techniques, formation evaluation, well completions and production, surface facilities, reservoir mechanics, offshore and onshore structures, environmental considerations; new technologies, prospects; petroleum industry activities in Lebanon and the Middle East; available industry and challenges; Oil & gas safety issues.		

APET 201	**Crude Oil & Natural Gas Transportation & Storage**	**3(3,0)**
This course will introduce the petroleum student to natural gas and crude oil transportation and storage equipment design and operation; following the flow of gas from the wellhead to the gas & oil pipelines, to the storage tanks; calculations are performed on pipeline sizing, inspection, and installation; storage tank design, installation, and inspection; use of commercial software. **Prerequisite**: APET 200.		

APET 203	**Modern Refining Technology**	**3(3,0)**
This course presents an introduction to petroleum refining technology, an overview of crude oil supply and petroleum product demand, and a description of refinery process technology. Some refining technologies are described, such as crude oil distillation, heavy oil conversion options, hydrotreating, and catalytic reforming. Some gas processing is also discussed. **Prerequisite**: APET 200.		

CHEM 213	**Physical and Organic Chemistry**	**3(3,0)**
This course will allow students to learn chemical principles related to the general, Physical, and Organic fields of chemistry. Topics include study of different states of matter, homogeneous and heterogeneous mixtures, study of endothermic and exothermic processes, discussion on surface phenomena and industrial applications of adsorption and absorption processes; Structure and properties of organic compounds, synthesis and reactions of various compounds, mechanism of radical substitution, electrophilic addition and electrophilic aromatic substitution, Liquids and liquid-gas solutions, viscosity and surface tension of liquids, name and write formulas		

of selected organic compounds generally found in oil and gas industries; Catalyst types and use. **Prerequisite**: general or environmental chemistry.

GEOL 210	**Petroleum Geology**	**3(3,0)**
This course introduces the geological concepts important to petroleum exploration and exploitation. Topics include the composition of the Earth, plate tectonics, rocks and minerals, the formation of sediments and sedimentary rocks, stratigraphy, geologic structures, oil and gas sources, reservoir engineering properties, exploration techniques, well evaluation, and reservoir development.		

II. Elective Course

Students may choose one three-credit course from the following list.

APET 202	**Materials Selection – Corrosion Monitoring and Control**	**3(3,0)**
Topics will provide the student with an understanding of the factors involved in materials utilization in petroleum equipment design and construction, such as pressure vessels & piping, materials properties, basic welding principles, and the applicable codes, standards, and engineering specifications used for materials specification in the hydrocarbon industries. Common non-destructive testing procedures used on equipment include identifying defect types, such as corrosion and cracks. Equipment strategy: monitoring and control. **Prerequisite**: APET 200.		

APET 300	**Applied Fluid Mechanics**	**3(3,0)**
Review of incompressible flow, turbulence, and dimensional analysis; pipe flows; boundary layers, lift, drag, and flow separation; compressible flow; turbomachinery and rotating machinery; oil and gas issues.		

III. Program Prerequisites

Students who wish to study the APET minor must complete the following courses.

Course Code	Title	Credits
CHEM 211	Environmental Chemistry and Lab	3
ENGL 217	Professional English Communication	3
MATH 215	Calculus III	3
MATH 311	Linear Algebra with Applications	3
MATH 314	Ordinary Differential Equations	3
MATH 351	Probability and Statistics	3
MECH 320	Mechanics of Materials	3
MECH 330	Thermodynamics I	3
MECH 335	Fluid Mechanics	3
PHYS 210	Fundamentals of Physics and Lab	3